

Critical Thinking questions

1.

a) Four legal identifier names

- **userName**
- **_index**
- **totalAmount**
- **age**

b) Four illegal identifier names with explanation

- **2ndNumber** (cannot start with a digit),
- **my var** (Cannot contain spaces),
- **int** (cannot use reversed keywords),
- **user-name** (Cannot contain invalid special characters such as "-")

2.

a) Declare a variable and assign a value in 2 statement

int numBeads;

numBeads = 5;

b) Declare a variable and assign a value in 1 statement

int numBeads = 5;

3. Final value of *yourNumber* after the last statement?

a) **13**

Final value of *yourNumber* after the last statement?

b) **11**

4. Determine appropriate data types...

- a) int
- b) double
- c) int
- d) double
- e) boolean
- f) character (char)

5. What is the difference between a primitive data type and an abstract data type?

- a) - **Primitive data types** are the most basic types of data built directly into a programming language. They store a single, simple value.

- **Abstract data type** is a type that is defined by: what it can do, what kind of data it stores, and is NOT defined by how it is built.

b) What is the difference between a class and an object?

Class - Is a blueprint or template, it defines what data something has, and what actions it can perform.

Object - Is an actual instance of a class, it stores real value, and can use the methods defined in the class.

11. Rewrite the following statements to include the appropriate type casting, rounding, if type casting is not needed explain why:

- a) $y = (\text{int})(j * k);$ (Because you're assigning a *double* to an *int*)
- b) $z = j * k;$ (No casting needed, result is already a *double*)
- c) $z = k * k;$ (No casting needed, result is already a *double*)
- d) $j = (\text{int}) k;$ (You must cast because you're converting from *double* to *int* this cuts off the decimal)
- e) $k = j;$ (No cast needed, this is automatically widening conversion)*****
- f) $y = j + 3;$ (No casting needed, everything is already *int*)